

Mattia Sospetti

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AI Engineer specialized in agentic and GenAI systems on AWS, from PoCs to production. Currently automating insurance claims handling for Stellantis Financial Services, with a background in computer vision, diffusion models and applied research.

Experience

Data Reply France - AI Engineer

September 2025 - Present (Paris, France)

Stellantis Financial Services - Claim Handling Efficiency,

- Built and led to production an agentic system automating insurance claims handling (**245 claims/month**), running a multiagentic architecture (Claude Sonnet 4.6, Strands, AgentCore), providing document intelligence + automated decisioning.
- Reached **97.3% classification accuracy**, **99.4% field extraction**, and **96.4% check accuracy** across 84 automated controls, including vision-based checks catching tampered documents.
- Optimized latency to ~106s/63s per claim at **\$0.53/claim**; added production-grade LLMOps, observability, monitoring, and canary deployment.

Primo - AI-powered IT Support Copilot

- Built a serverless Level 1 support agent (MCP server, 7+ tools, AgentCore Gateway) with multi-source RAG via Tavily; **10-12s response time**, **\$0.19/conversation**, 24/7 multilingual.

Prismic - AI Content Optimization PoC

- Built a 4-agent pipeline (with LLM-as-judge scoring) to optimize web content for SEO/GEO against brand voice; **+50% SEO/GEO score**, sub-1-minute turnaround.

GrottiniLab - Machine Learning Engineer

May 2023 - September 2025 (Macerata, Italy)

- Improved reverse image search accuracy from **92% to 96%** through focal loss, a newer Transformer architecture, and hyperparameter optimization, for a product optimizing supermarket supply chains.
- trained **diffusion models** (LoRA, ControlNet) for personalized product generation.
- Built an LLM-based natural-language interface to the data warehouse (text-to-SQL, **RAG** with router + vector DB) for non-technical querying.
- Cut embedded tracking unit cost from **€500 to €100** with custom firmware; ported the tracking stack to Python and scaled monitoring to 100 edge devices on AWS.
- Built a 360° remote monitoring system for construction sites (Jetson Nano, ZED stereo cameras) with real-time obstacle detection.

Télécom SudParis - Visiting Researcher Intern

April 2024 – May 2025 (Paris, France)

- Researcher on the **Trust Rise Project** (Marie-Curie fellowship), co-authoring R&D academic articles in AI and IoT security.
- Implemented a real-world simulation system to test **MQTT protocols** from a cybersecurity perspective.
- Developed **classification neural networks** whose accuracy increased by 10% through the adoption of more modern networks and improved training methods.
- Implemented a **backdoor injection** on a **GAN neural network** for risk analysis.
- Paper publication: <https://inria.hal.science/hal-04832442/>

Education

Università Politecnica delle Marche Computer Science and Automation Engineering (Master's Degree) — 110/110 cum laude

2017-2023

- Final Dissertation: “Deep Learning to intercept trends for a fashion company”

Certifications & Achievements

AWS Certified AI Practitioner • AWS Certified Solutions Architect – Associate • 3rd place, 2025 Paris GenAI Partner Hackathon

Skills

ML/AI: Python, SQL, PyTorch, TensorFlow, HuggingFace, LangChain, Strands Agents, OpenCV, Scikit-Learn, Pandas, RAG, agentic systems, diffusion models (LoRA/ControlNet), LLMOps

Cloud/Infra: AWS Bedrock & AgentCore, Textract, Lambda, DynamoDB, S3, RedShift, CloudWatch, Terraform, Docker, Step Function, Cloudwatch, Artifactory, TeamCity

Languages: Italian (native), French (fluent, C1 working proficiency), English (C1 certificate)